Data Mining and Discovery Project

UK Transport Network Database

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Introduction: this project is about a synthetic database model of UK Public Transport Network. The main aim is to build a strong and highly normalized schema to keep track train journeys, ticketing data, operator details, and station logistics which resulting SQLite database demonstrates relational design and data constraints and realistic data modeling techniques.

Database Creation: for data creation I written python script using data generation libraries to synthesize realistic transport records, I have built realism into the dataset. this is programmed as script to insert missing values null's into 5% of delay_minutes records in effort to simulate dropped missing updates. I created a staging table called raw_journey_import to ccopy an API ingestion point. I inserted duplicate rows in this table so the affecting records where journey_id % 40 == 0. This design forces downstream processes to implement SQL duplication techniques such as GROUP BY clauses window function to remove all duplicates for analytical pproces.

Schema and Table:

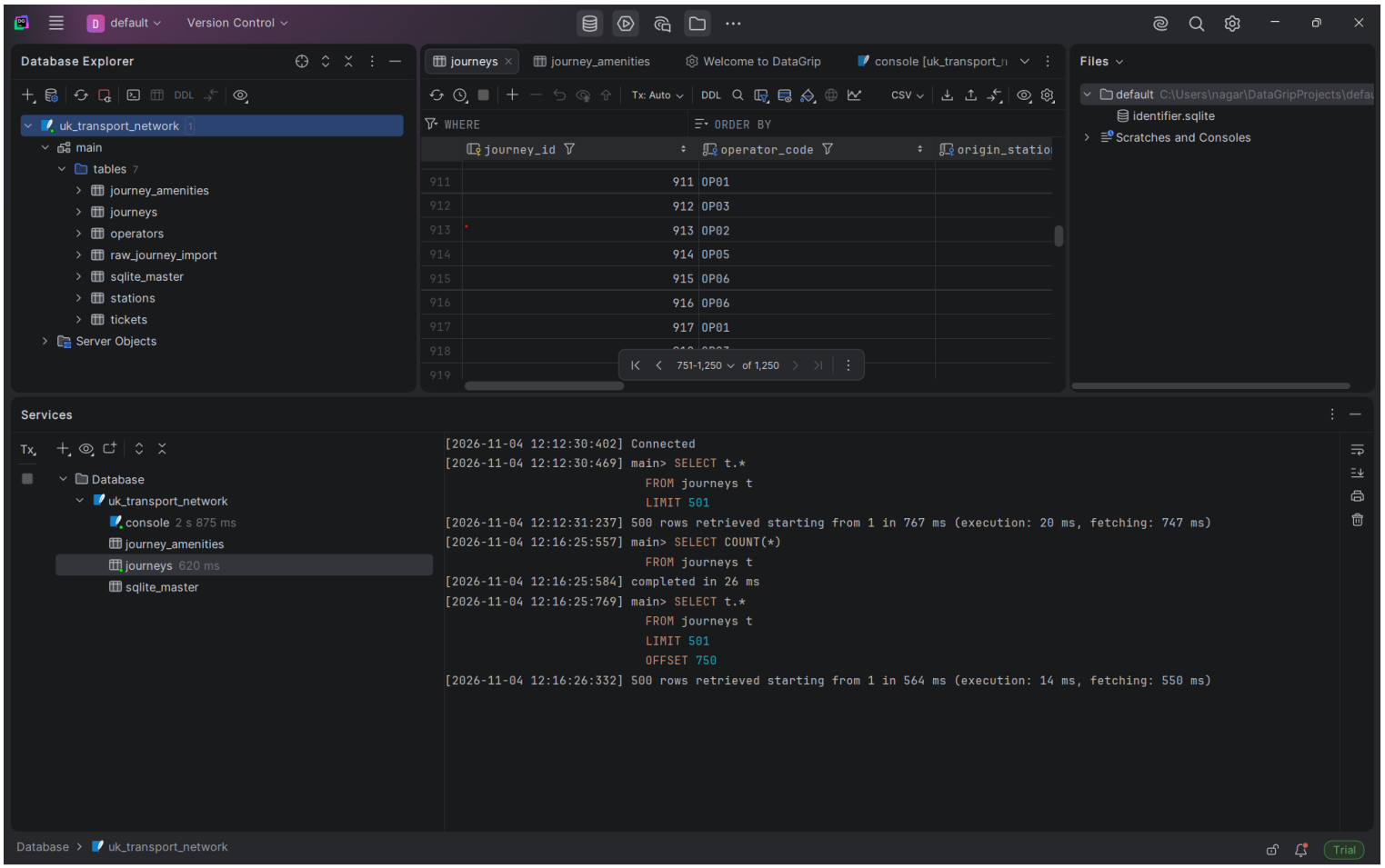


Figure 1: Database Table generation having 1000+ rows

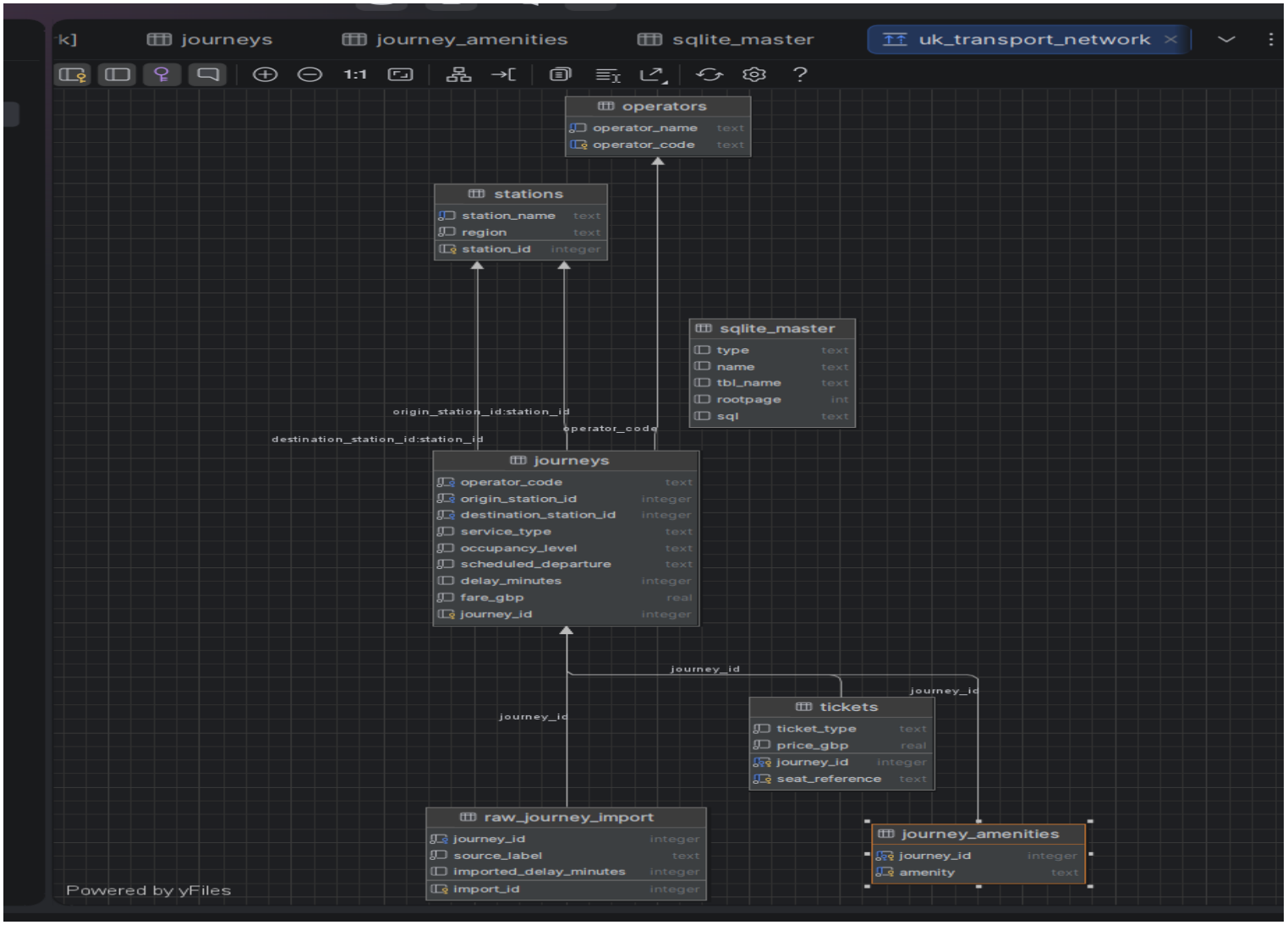


Figure 2: Database Entity Relationship diagram

The database is normalized with 6 different tables to keep integrity. Separation of these entities in the database helps to avoid redundancy and update anomalies .

journeys: This is the central transaction table where 1,250 records are present.

stations and operators: Lookup tables which keeps repeated station locations and operator names in one place.

tickets: Split from trips as a trip will have many tickets.

journey_amenities: Solves many to many relationship for multiple amenities such as Wi-Fi or Bike space to be logged per journey..

raw_journey_import: Raw and uncleaned telemetry data staging ground in an unstructured format .

Schema Links: The schema makes heavy use of strict relational constraints to enforce data integrity. The table journeys has two connections to stations via origin_station_id and destination_station_id and one to operators via operator_code. I have used composite primary keys in two different tables to avoid logical duplicates of tickets which uses journey_id, seat_reference so that same seat cannot be sold twice and journey_amenities uses the journey_id, amenity. Domain constraints are also hard coded and a CHECK the origin_station_id <> destination_station_id to prevents a train from traveling to its own origin, and CHECK the fare_gbp >= 0 which ensures prices cannot be negative.

Data Types and Variables:

Data Type	Columns	explanation
Nominal	service_type, region	Categorical data that are not inherently ordered numerically,for example, 'Intercity', 'Commuter'.
ordinal	occupancy_level	Ranked qualitative data where order is important such as 'Low', 'Medium', 'High', 'Full'.
interval	scheduled_departure	Datetime values where the distance between two points is meaningful but there is no absolute zero.
ratio	fare_gbp, delay_minutes	Quantitative data with a true zero point so that meaningful ratios can be calculated as 60 minute delay is twice as long as a 30 minute delay.

Conclusion: The UK Transport Network database presents realistic approach to data modelling . The system is reliable because it uses a normalized schema with strict integrity with multiple composite keys and cross column checks . The intentional injection of realistic data anomalies are missing delays and duplicated which import feeds shoes the need for robust data cleaning pipelines. The careful removing of ticketing records to underscores a commitment to data minimization principles with a balance between desire for analytical insights and ethical obligation to safeguard passenger privacy .

Link to code via github: https://github.com/Vishnuvardhan2503/Data-Mining-/upload/main